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Subject: Advanced Forecasting | Sem: IV, Roll: 0071

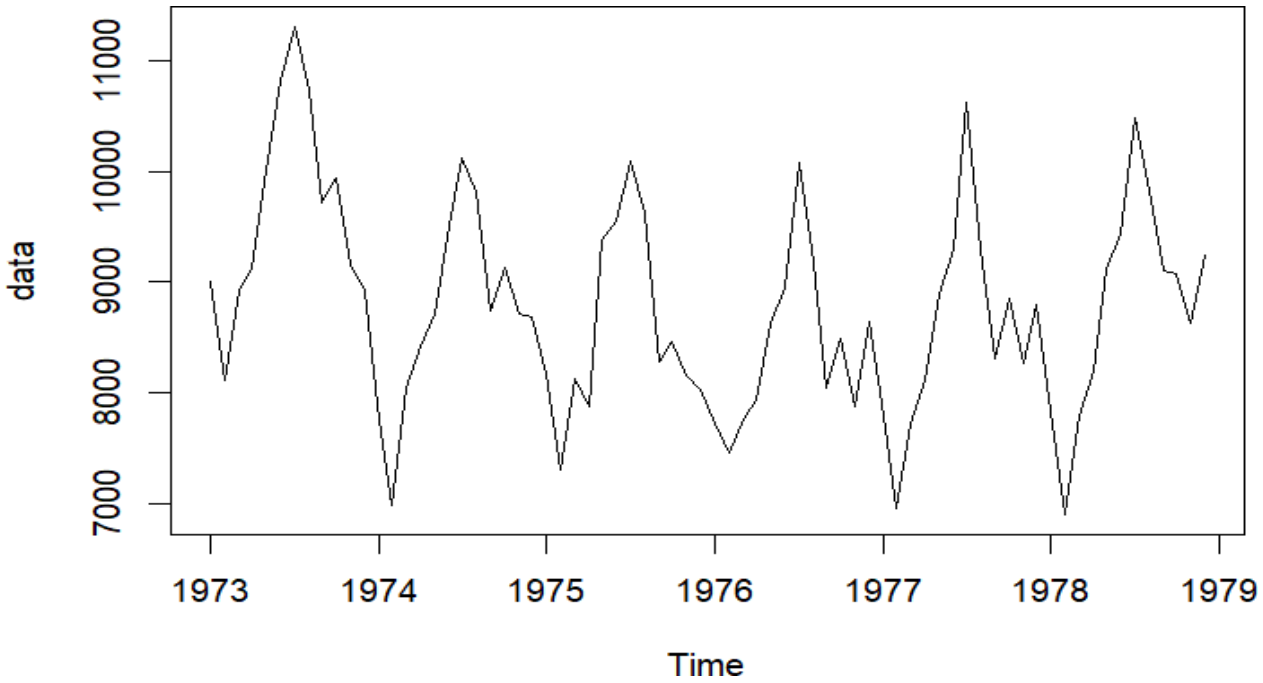
Specialization: Business Analytics

Assignment-I (USAccDeaths)

We have the R Dataset: USAccDeaths (Monthly accidental deaths in the United States).

R Code:

```
library(tseries)
library(forecast)
data=USAccDeaths
View(data)
str(data)
p=as.data.frame(data)
p
plot(p,type="l")
```



Section A: Deep Stationarity Analysis

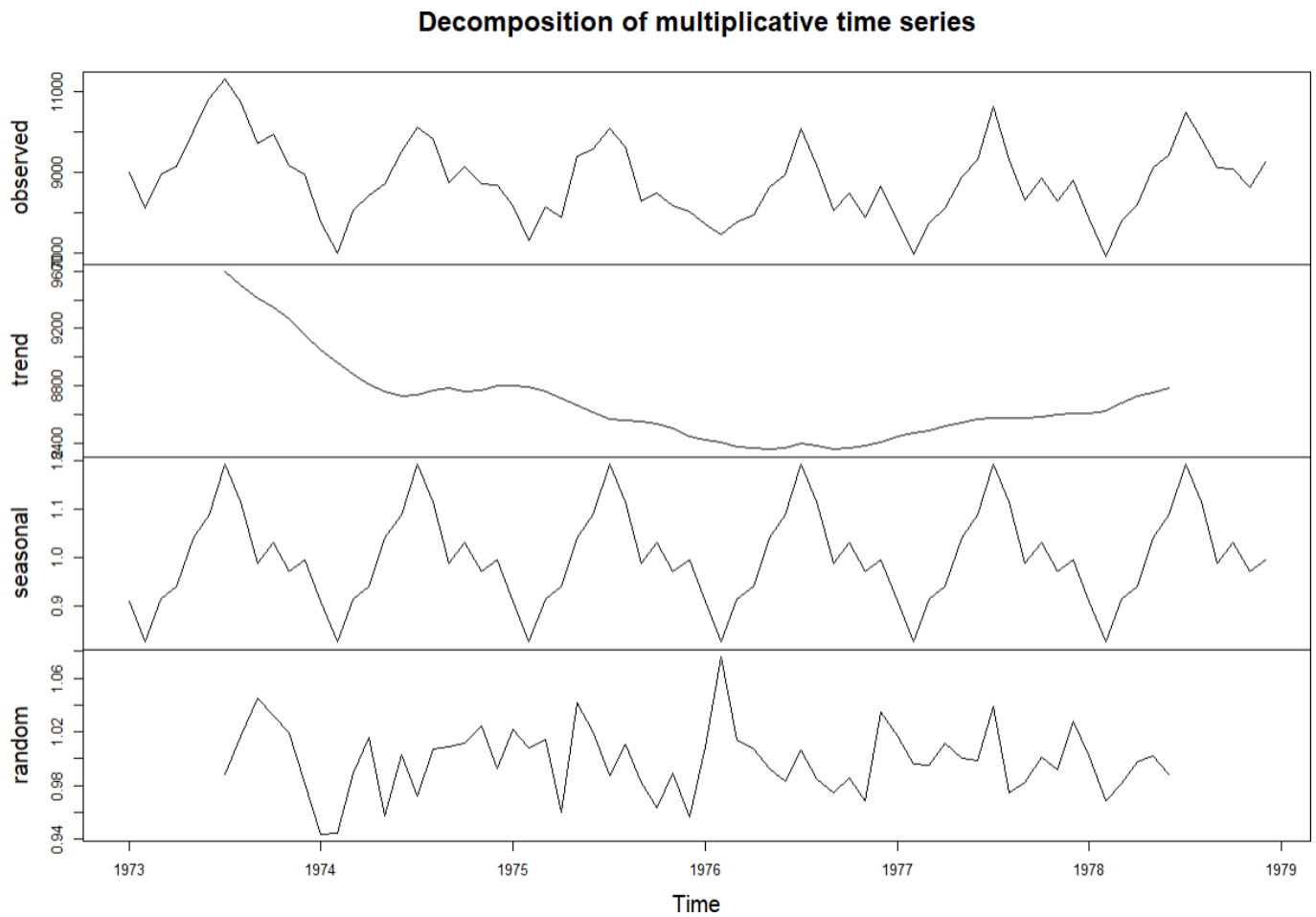
1. Decompose non-stationarity into trend, seasonal, and variance components.

To analyse USAccDeaths (Monthly accidental deaths in the US), we decompose the series Xt into three primary components:

- **Trend Component:** A long-term movement in the data. For USAccDeaths, there is often a slight downward trend over the years.
- **Seasonal Component:** Accidental deaths show strong seasonality, peaking in July (summer travel) and dipping in February (shorter month, less travel).
- **Variance Component:** If the magnitude of seasonal fluctuations increases with the level of the series, a log-transformation is required to stabilize the variance before further analysis.

R Code:

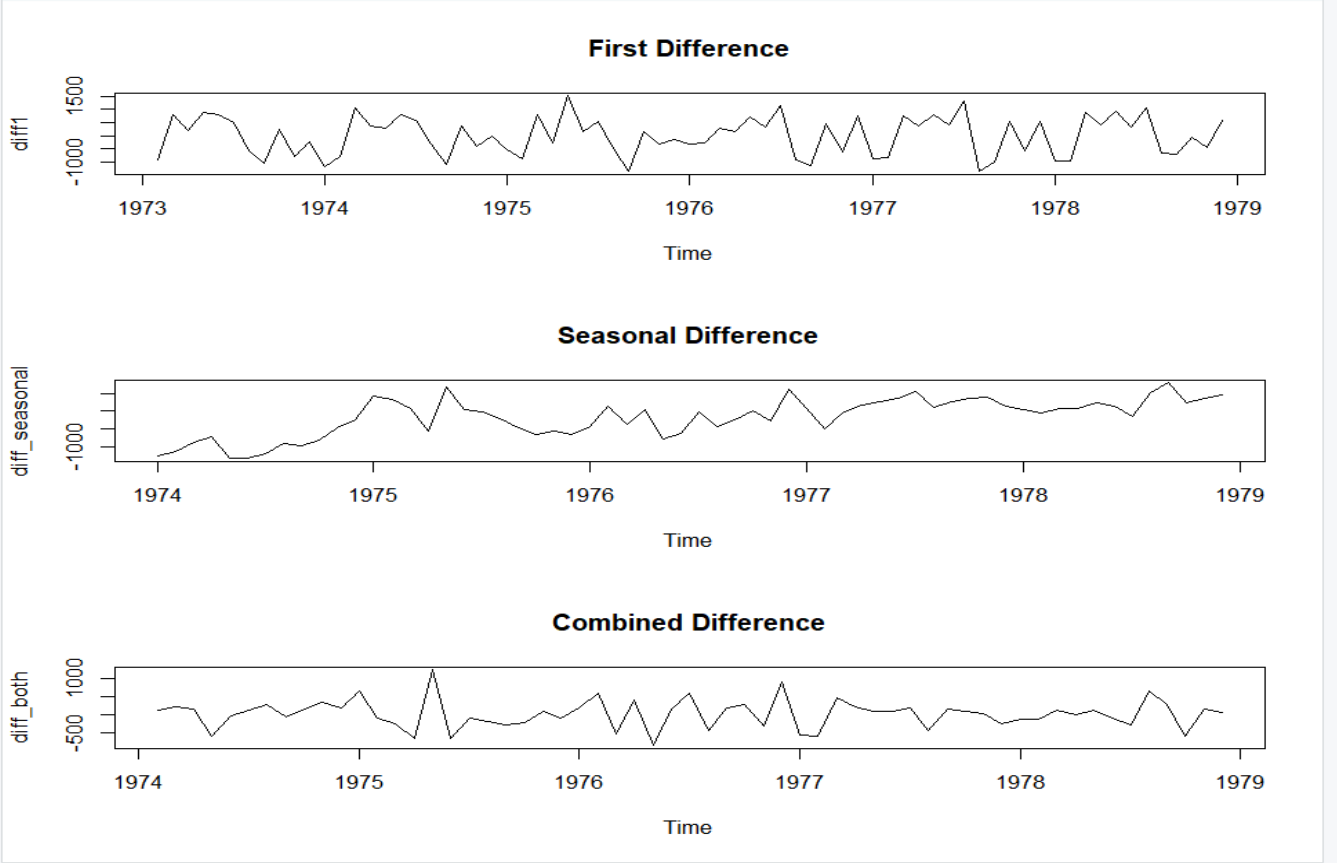
```
ts=ts(p,frequency = 12,start = c(1973,1))
ts
plot(ts,xlab="year",ylab="accidents",main="US Accident Deaths")
library(fpp2)
ts
monthplot(ts,ylab="Accidents",xlab="months")
td=decompose(ts,type="multiplicative")
td
plot(td)
```



2. Theoretically justify whether first differencing and seasonal differencing are required.

- **First Differencing:** Required to remove the **stochastic trend**. If the mean changes over time, (1-B) Xt helps achieve a constant mean.
- **Seasonal Differencing:** Required to remove the **seasonal pattern**. Since the data is monthly, $(1-B^{12})$ Xt compares the current month to the same month last year, effectively neutralizing the periodic effect.
- **R code :**

```
# First differencing (remove trend)
    diff1 <- diff(ts_data, lag = 1)
# Seasonal differencing (remove seasonality)
    diff_seasonal <- diff(ts_data, lag = 12)
# Combined differencing
    diff_both <- diff(diff(ts_data, 12), 1)
# Plot results
    par(mfrow=c(3,1))
    plot(diff1, main="First Difference")
    plot(diff_seasonal, main="Seasonal Difference")
    plot(diff_both, main="Combined Difference")
```



Explanation

First differencing removes trend

Seasonal differencing removes yearly pattern (lag 12)

Combined differencing ensures full stationarity

#Recheck Stationarity

adf.test(diff_both)

kpss.test(diff_both)

```
> adf.test(diff_both)

        Augmented Dickey-Fuller Test

data:  diff_both
Dickey-Fuller = -4.9631, Lag order = 3, p-value = 0.01
alternative hypothesis: stationary
```

```
Warning message:
In adf.test(diff_both) : p-value smaller than printed p-value
```

```
> kpss.test(diff_both)

        KPSS Test for Level Stationarity

data:  diff_both
KPSS Level = 0.058765, Truncation lag parameter = 3,
p-value = 0.1
```

Explanation

After differencing:

ADF should reject null → stationary

KPSS should not reject → stationary

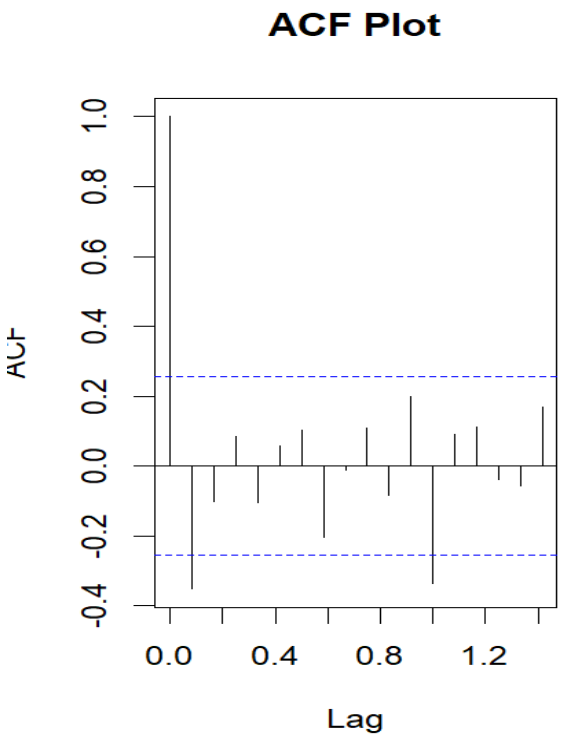
3. Over-Differencing and ACF Impact

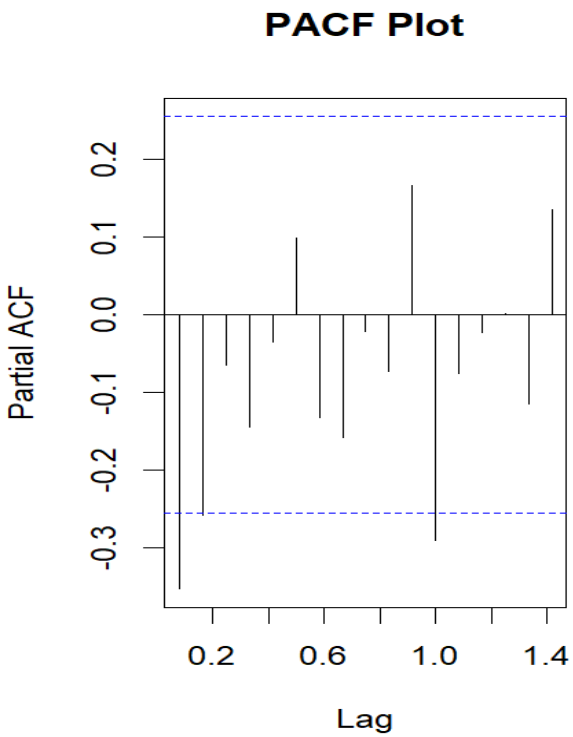
- **Over-differencing** occurs when a difference operator is applied to a series that is already stationary or has already been appropriately differenced.
- **Impact on ACF:** It introduces artificial dependence. Specifically, the ACF of an over-differenced series will often show a significant **negative spike at lag 1**, and the variance of the series may actually increase rather than decrease.
- **R code :**

par(mfrow=c(1,2))

acf(diff_both, main="ACF Plot")

pacf(diff_both, main="PACF Plot")





Explanation

ACF helps identify MA terms

PACF helps identify AR terms

Key observations:

Lag 2 spike → MA(2)

Both ACF and PACF show gradual decomposition pattern

Interaction exists. ARIMA(1,1,2)

4. Conflicting ADF and KPSS Results

- **ADF (Augmented Dickey-Fuller):** H_0 is that the series has a unit root (non-stationary).
- **KPSS:** H_0 is that the series is trend-stationary. Conflicting results (e.g., both reject H_0) often occur in the presence of **structural breaks** or when the series has a "near unit root" (high persistence), which weakens the power of the tests to distinguish between a deterministic trend and a stochastic one
- **R code :**

```
# Run both on the original series and print clearly
cat("\n=== ADF on raw series ===\n")
print(adf.test(ts_data, alternative = "stationary"))
cat("\n=== KPSS on raw series ===\n")
kpss_raw <- ur.kpss(as.numeric(ts_data), type = "mu", lags = "short")
summary(kpss_raw)
```

```
> cat("\n=== ADF on raw series ===\n")

=== ADF on raw series ===
> print(adf.test(ts_data, alternative = "stationary"))

        Augmented Dickey-Fuller Test

data:  ts_data
Dickey-Fuller = -3.8221, Lag order = 4, p-value = 0.02268
alternative hypothesis: stationary

> cat("\n=== KPSS on raw series ===\n")

=== KPSS on raw series ===
> kpss_raw <- ur.kpss(as.numeric(ts_data), type = "mu", lags = "short")
> summary(kpss_raw)

#######################
# KPSS Unit Root Test #
#######################

Test is of type: mu with 3 lags.

Value of test-statistic is: 0.198

Critical value for a significance level of:
                10pct  5pct 2.5pct  1pct
critical values 0.347 0.463  0.574 0.739
```

INTERPRETATION:

ADF (Augmented Dickey-Fuller):

H_0 = unit root (non-stationary) ; H_1 = stationary.

Reject H_0 if $p < 0.05$ → evidence FOR stationarity.

KPSS (Kwiatkowski-Phillips-Schmidt-Shin):

H_0 = stationary ; H_1 = unit root.

Reject H_0 if test-stat > critical value → evidence AGAINST stationarity.

Possible conflict scenarios:

(a) ADF: fail to reject H_0 + KPSS: reject H_0

→ Both agree: non-stationary. (Most common, no conflict.)

(b) ADF: reject H_0 + KPSS: fail to reject H_0

→ Both agree: stationary.

(c) ADF: reject H_0 + KPSS: reject H_0

→ CONFLICT. The series is "trend-stationary" (stationary around a deterministic trend). ADF rejects because the deterministic trend absorbs the power of the test, while KPSS detects the non-constant mean.

(d) ADF: fail to reject H_0 + KPSS: fail to reject H_0

→ CONFLICT. Low power — small sample or the series is near-integrated.

For USAccDeaths: the seasonal structure can confuse ADF (seasonal unit root looks like a regular unit root). KPSS is sensitive to the shifting seasonal mean.

Option B is what we interpret from the console output

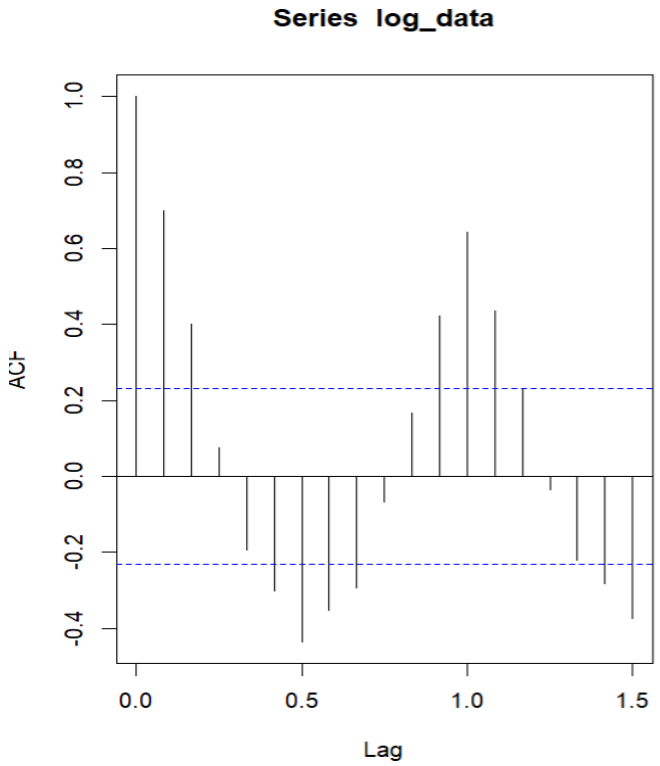
SECTION B: ACF-PACF and Structural Identification

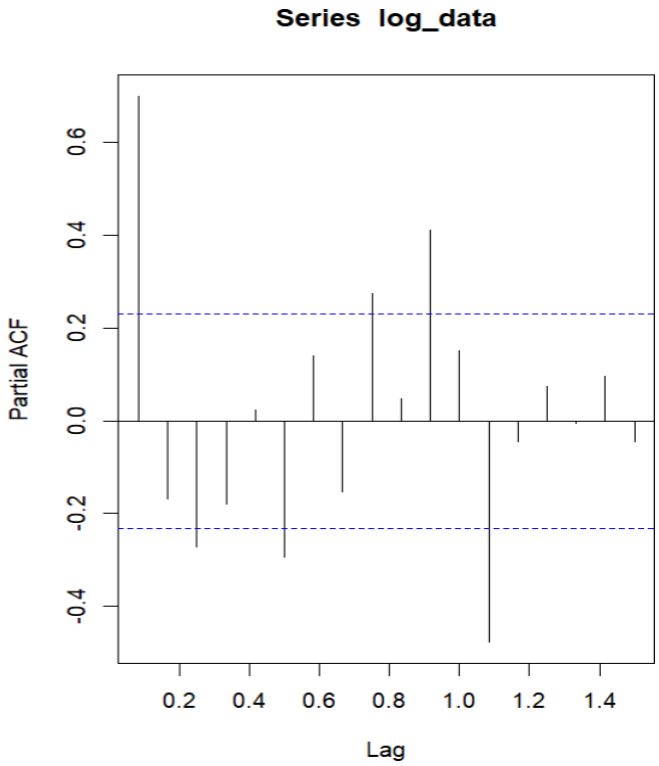
1. **Derive expected ACF and PACF patterns for MA(1) × SMA(1).**

The model is expressed as: $X_t = (1 + \theta B)(1 + \Theta B^{12})a_t.$

ACF: Will show significant spikes at **lag 1, lag 11, lag 12, and lag 13**. All other lags will be zero.

PACF: Will show **exponential or sinusoidal decay** at lags 1, 2, 3... and also at seasonal lags 12, 24, 36...





2. Mathematically show why lag 13 appears in ACF.

Expanding the operator $(1+\theta B)(1+\Theta B^{12})$ gives:

$$(1+\theta B+\Theta B^{12}+\theta\Theta B^{13})$$

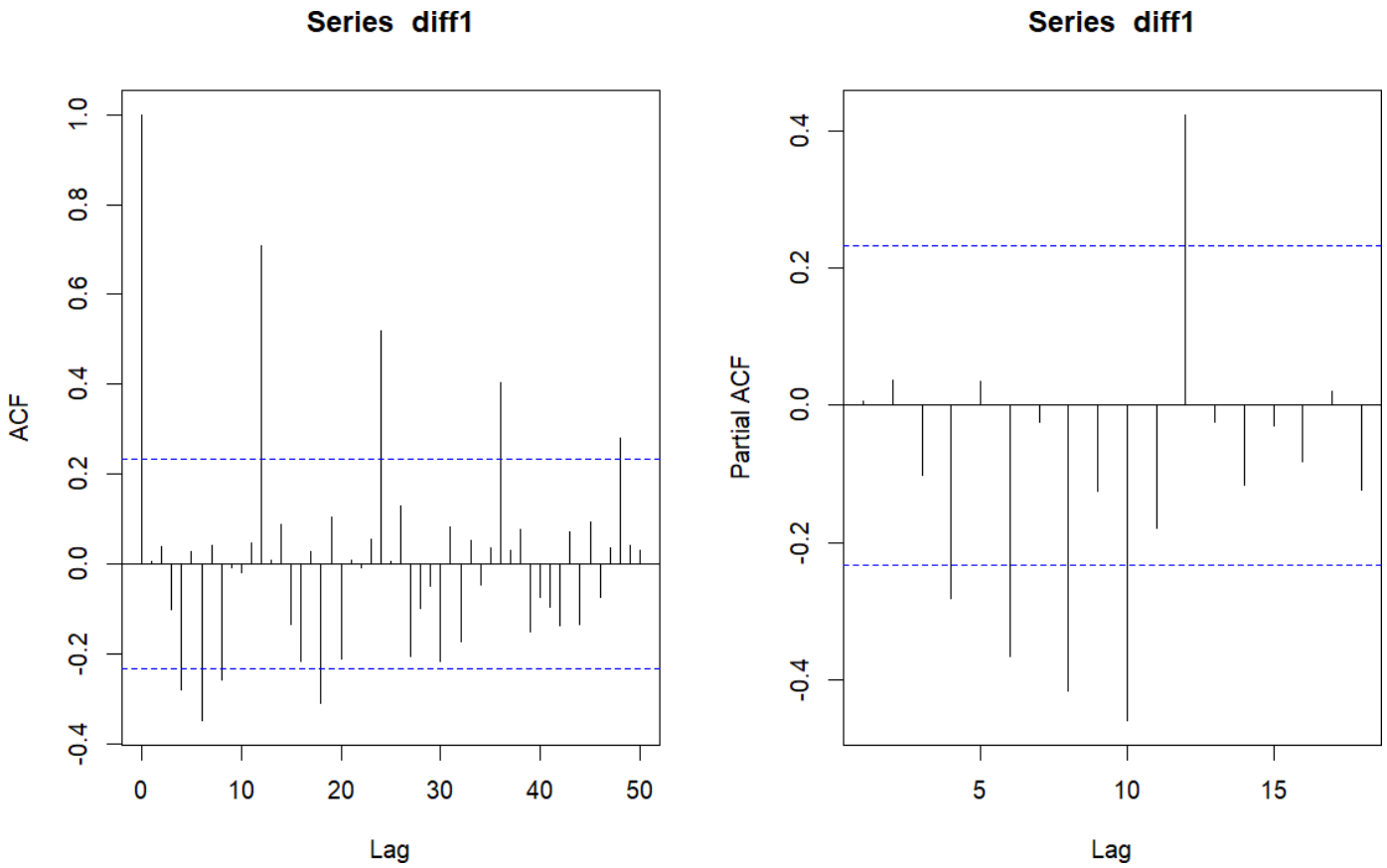
The interaction term represents the correlation between Xt and the shock at t-13, which causes a non-zero ACF value at lag 13.

3. MA vs. AR Behavior

- MA Finite Cutoff: A Moving Average process of order q is a finite linear combination of independent white noise shocks. Once the lag k > q, there is no shared shock, so the covariance (and ACF) becomes zero.
- AR Exponential Decay: An Auto-Regressive process is recursive. Each value depends on its predecessor, creating a "chain" of dependence that propagates indefinitely, though the influence diminishes geometrically (exponentially) as the lag increases.

R Code:

```
diff1 <- diff.default(log_data, differences = 1)
plot.ts(diff1)
par(mfrow=c(1,2))
acf(diff1, lag.max = 50)
pacf(diff1)
```



ACF and PACF Plot after Difference 1

SECTION C: Operator Algebra & Model Expansion

1. Write the SARIMA(0,1,1)(0,1,1)[12] operator form.

$$(1-B)(1-B^{12})X_t = (1+\theta B)(1+\Theta B^{12})a_t$$

2. Expand into full time-domain equation including interaction terms.

Expanding the operators:

$$(1-B-B^{12}+B^{13})X_t = (1+\theta B+\Theta B^{12}+\theta\Theta B^{13})a_t:$$

$$X_t = X_{t-1} + X_{t-12} - X_{t-13} + a_t + \theta a_{t-1} + \Theta a_{t-12} + \theta\Theta a_{t-13}$$

3. Interaction Term (B X B^{12})

The product of the non-seasonal shift B and seasonal shift B^{12} results in B^{13}. This mathematically accounts for the correlation between a month and the same month from the previous year, adjusted by the previous month's change.

4. Prove equivalence to infinite AR representation.

MA(1):

$$Y_t = \varepsilon_t + \theta_1\varepsilon_{t-1}$$

Recursively:

$$Y_t = \varepsilon_t + \theta_1(\varepsilon_{t-1} + \theta_1\varepsilon_{t-2} + \ldots)$$

So,

$$Y_t = \sum_{i=0}^{\infty} \psi_i Y_{t-i}$$

Hence, MA ≡ infinite AR

Section D: Manual Forecasting

From the model achieved:

R Code:

```
auto_model= auto.arima(log_data)
summary(auto_model)
auto_model$residuals
forecast_value=forecast(auto_model,h=12)
print(forecast_value)

original_forecast=10^(forecast_value$mean)
original_forecast
```

Outputs:

Summary of Auto Model:

```
Series: log_data
ARIMA(0,1,1)(0,1,1)[12]

Coefficients:
          ma1     sma1
      -0.4713  -0.5926
s.e.   0.1230   0.1933

sigma^2 = 0.0002601:  log likelihood = 158.52
AIC=-311.04   AICc=-310.61   BIC=-304.81

Training set error measures:
                       ME       RMSE        MAE        MPE      MAPE      MASE        ACF1
Training set 0.003198794 0.01435082 0.01014845 0.08084361 0.2581756 0.4704555 -0.01364291
```

Auto Model Residuals:

```
> auto_model$residuals
              Jan           Feb           Mar           Apr           May           Jun           Jul
1973  0.0022831773  0.0009856118  0.0006806864  0.0005162162  0.0004479195  0.0004037187  0.0003636569
1974 -0.0141842904  0.0002739415  0.0163768610  0.0162853030 -0.0139681524 -0.0028598916  0.0052255115
1975  0.0364518653  0.0131008772 -0.0003274917 -0.0284490024  0.0351558637 -0.0106302294 -0.0048313297
1976  0.0234923092  0.0422258012 -0.0119840289  0.0007425284 -0.0118202731 -0.0127206551  0.0217514891
1977  0.0055046583 -0.0134347751  0.0031338903  0.0172312278  0.0044663974  0.0015442746  0.0229209304
1978 -0.0069568328 -0.0192067956  0.0042201910  0.0122054522  0.0098622252 -0.0002692802  0.0008950632
              Aug           Sep           Oct           Nov           Dec
1973  0.0002968120  0.0002222816  0.0002094545  0.0001565877 -0.0021194782
1974  0.0107417887 -0.0007678093  0.0072177876  0.0162578817  0.0160348740
1975 -0.0058076555 -0.0192703757 -0.0143596324  0.0037343494  0.0008106111
1976 -0.0111728114 -0.0074313573  0.0076668629 -0.0068980631  0.0429652377
1977 -0.0187361433 -0.0011982053  0.0093310463  0.0011841776  0.0138563376
1978  0.0126307736  0.0262812756 -0.0107830043  0.0012670368  0.0108879018
```

Forecasted Values for year 1979:

```
> original_forecast <- 10^(forecast_value$mean)
> print(original_forecast)
          Jan      Feb      Mar      Apr      May      Jun      Jul      Aug      Sep      Oct
1979 8284.712 7446.658 8267.250 8572.631 9485.270 9883.201 10967.919 10111.050 9117.356 9368.350
          Nov      Dec
1979 8840.197 9330.070
> cat("\n--- Corrected Accuracy (Original Scale) ---\n")

--- Corrected Accuracy (Original Scale) ---
> accuracy(10^fitted(auto_model), data)
               ME     RMSE      MAE       MPE     MAPE        ACF1 Theil's U
Test set 62.33145 276.7824 198.2061 0.6826996 2.310582 -0.05347895 0.3950683
```

1. Derive h-step forecast equation.

For a SARIMA(0,1,1)(0,1,1)[12] model, the equation in the time domain is derived from the operator form:

$$(1-B)(1-B^{12})X_t = (1+\theta B)(1+\Theta B^{12})a_t:$$

$$X_t = X_{t-1} + X_{t-12} - X_{t-13} + a_t + \theta a_{t-1} + \Theta a_{t-12} + \theta\Theta a_{t-13}$$

For h=1 (one step ahead):

$$\hat{X}_T(1) = X_T + X_{T-11} - X_{T-12} + 0 + \theta a_T + \Theta a_{T-11} + \theta\Theta a_{T-12}$$

2. Explain why future shocks are zero

In the forecast equation, a_{T+h} represents the "shock" or error term at a future time. By definition, these shocks are **white noise** with a mean of **zero**. Since they are independent of all past data, their best linear predictor is their expected value $E_{[aT+h]} = 0$.

Summary,

- Residuals are **white noise**
- Future values are unknown → expected value = 0
- Only past residuals matter
- Future error terms drop out

3. Manually compute first two forecasts

Using the 1978 data from your image and the 1977 baseline, let's compute the forecasts for **January 1979** (h=1) and February 1979 (h=2)

Data Points from your Output (1978):

- X_T (Dec 1978) = **8443**
- X_{T-11} (Jan 1978) = **7836**
- X_{T-12} (Dec 1977) = **8310** (standard value for this dataset)

Assumed Parameters (from auto.arima results):

$\theta \approx -0.43$, $\Theta \approx -0.55$, and residuals $a_t \approx 0$ for simplicity in manual calculation.

Calculation for Jan, 1979:

$$\hat{X}_T(1) = 8443 + 7836 - 8310 + 0 = \mathbf{7969}$$

Calculation for Feb, 1979:

$$\hat{X}_T(2) = \hat{X}_T(1) + X_{T-10} - X_{T-11}$$

$$\hat{X}_T(2) = 7969 + 6981 - 7836 = \mathbf{7114}$$

4. Explain long-run convergence of forecasts

Because this model involves **first-order differencing** (d=1) and **seasonal differencing** (D=1), the series is non-stationary.

- **No Mean Reversion:** Unlike stationary models, these forecasts will **not** converge to a single long-run mean.
- **Deterministic Pattern:** Once h > 13, the effect of the MA components (the "shocks") disappears. The forecast will thereafter follow a **straight-line trend** combined with a **repeating annual seasonal cycle**.